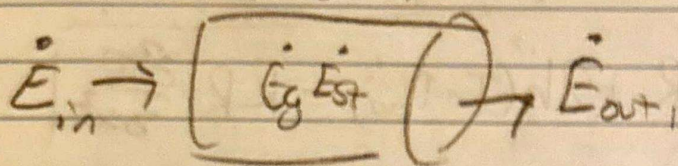


Hw3 Robert Brown

1) 1D hg w/ heat generation.

$$k \frac{\partial^2 T}{\partial x^2} + \dot{q} = 0$$

 $\leftarrow \partial x \rightarrow$


a)

$$\dot{E}_{st} = \dot{E}_g + \dot{E}_{in} - \dot{E}_{out,1} - \dot{E}_{out,2}$$

$$\begin{aligned} \dot{E}_{st} &= \rho V c \frac{\partial T}{\partial t} = \rho \pi R^2 \Delta x c_p \frac{\partial T}{\partial t} \\ &= \rho \cdot \pi R^2 \cdot c_p \cdot \frac{\partial T}{\partial t} \cdot \Delta x \end{aligned}$$

$$\dot{E}_g = \dot{q} \pi R^2 \Delta x$$

$$\begin{aligned} \dot{E}_{out,1} &= h \pi R^2 (T - T_\infty) \cdot \Delta x \\ \dot{E}_{out,2} &= 2h \pi R (T - T_\infty) \cdot \Delta x \end{aligned}$$

$$\dot{E}_{in} - \dot{E}_{out,1} = -k \pi R^2 \frac{\partial^2 T}{\partial x^2} \Delta x$$

$$\dot{E}_{in} =$$

$$b) \rho R^2 c_p \frac{\partial T}{\partial t} = \dot{q} R^2$$

$$\text{So } \rho R^2 c_p \frac{\partial T}{\partial t} = \dot{q} R^2 + 2h\pi R(T - T_{\infty}) - k R^2 \frac{\partial^2 T}{\partial x^2}$$

$$\left| \rho c_p \frac{\partial T}{\partial t} = \dot{q} R + 2h(T - T_{\infty}) - k \frac{\partial^2 T}{\partial x^2} \right|$$

$$c) \quad T(x, t) \quad 0 \leq x \leq L$$

$$T(0, t) = T_i$$

$$T(x, 0) = T_i$$

$$T(L, t) =$$

not sure how
to deal w/ the
convective boundary
condition

$$-k \frac{\partial T}{\partial x} \bigg|_{x=L} = h(T_{\infty} - T(L, t))$$